

Music-Gen v3 SPINE Milestone — Cycles 17-19

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Music-Gen v3 SPINE Milestone — Cycles 17-19

Abstract

This report covers three consecutive heartbeat cycles of the M-V3-SPINE milestone on the reference track Chicken Grease (source SHA-16 31a164f845f8e27e). All three cycles run under the wait-on-operator cadence policy landed in Cycle 8 and neither of that policy’s two break-glass triggers — a live-guidance operator directive or an auditor CRITICAL finding — fires at any point in the arc. The three cycles land the policy’s prescribed four-slot shape three more times, extending the campaign’s consecutive-heartbeat streak from eight (through Cycle 16) to eleven and its clean-audit streak from twelve (through Cycle 16) to fifteen. Every earlier verdict remains byte-identical on disk; the anchor snapshot grows from 186 at Cycle 16 through 196, 206, and 216 across this arc at a steady +10 per cycle; the three-way rubric-v2 integrity chain, the thirteen-cycle venv byte-identity, the append-only verdict chain, and the durable Mode-2 lock on the torch-2.13 reproduction all remain intact. The one novel observation of the arc is a small elevation in the `promise_check` warning-count growth trajectory at Cycle 19 (+5 delta versus the previous +2/+3 envelope), logged by the auditor as MINOR and characterized as routine orphan-report accretion rather than a discipline drift. M-V3-SPINE-1 remains gated on operator ear per Fixed Decision 6; all downstream milestones remain frozen.

1. Continuity from Cycles 14-16

The Cycles 14-16 arc ended with a first fully-clean heartbeat audit (Cycle 16 was 0 CRITICAL / 0 MODERATE / 0 MINOR), closing the promise-check bookkeeping variance that had persisted across Cycles 12-15. The three cycles reported here — Cycles 17, 18, and 19 (the ninth, tenth, and eleventh consecutive heartbeats in the campaign) — repeat the heartbeat shape three more times without narrative change, extending the arithmetic invariants (venv chain, anchor growth, cycles-since-operator counter) by one segment per pass.

2. Methodology carried across all three cycles

The methodology is unchanged from the two prior arcs. Every new script sets the pinned environment (PYTHONHASHSEED=0, SOURCE_DATE_EPOCH=1756463424, TZ=UTC, LC_ALL=C.UTF-8, single-threaded BLAS via OMP/MKL/OPENBLAS_NUM_THREADS=1); every top-level invocation uses /usr/bin/python3 under an AST-verified interpreter guard; every new script is AST-scanned to fail on any network or PRNG import. The three continuing integrity chains all hold across the arc: rubric-v2 (document c49db5a12e955f26...451a == pinned hash file == each cycle's verdict.rubric_hash_v2), cadence policy (0be540365c8c03ad...c7f2 == its pinned hash file), and Cycle 7 torch-probe module (b54adadd...9af3d, byte-identical). The backref pattern continues: each cycle's verdict pins the previous cycle's verdict.json SHA in c<N-1>_backref_sha resolved from disk at emit time and re-verified by the auditor's walker test.

3. Heartbeat shape (identical across Cycles 17, 18, 19)

Each cycle delivers four artifacts, four housekeeping ledger rows, and a twelve-case test suite.

3.1 Track 1 — Torch-213 dry-run liveness roll-forward

The per-cycle caller (scripts/v3_spine/torch213_reproduce_probe_c<N>.py) imports the read-only Cycle 7 probe module and runs Mode 1 only, asserting the four baseline properties against a cumulative chain that extends by one segment each cycle:

Cycle	Baseline chain	attribution_verdict
17	vs c7..c16	ENV_DRIFT_PROBE_CANDIDATE_ FOUND_C17_DRY_RUN_ROLL_FORWARD
18	vs c7..c17	ENV_DRIFT_PROBE_CANDIDATE_ FOUND_C18_DRY_RUN_ROLL_FORWARD
19	vs c7..c18	ENV_DRIFT_PROBE_CANDIDATE_ FOUND_C19_DRY_RUN_ROLL_FORWARD

All four asserted properties — torch.__version__ = 2.13.0+cpu, torch.__file__ = /usr/local/lib/python3.11/dist-packages/torch/__init__.py, drafted reproduction command (binary form and -m muscriptor.cli module form) byte-identical to the Cycle 7 pin, and venv (workspace/learned_transcribers_venv/) directory-manifest SHA a86205175728d58f0a96ad02fc1ab1ac9e35f06c5ed568a960ed1ff261f83a74 — passed at every cycle. network_syscall_attempted=false and venv_signature_pre == venv_signature_post at every cycle. Mode 2 remains awaiting_operator_green_light; the durable Cycle 7 lock has now held across the thirteen cycles from Cycle 7 through Cycle 19.

The Cycle 19 probe carried the twelve-cycle inner key venv_manifest_matches_c7_c8_c9_c10_c11_c12_c13_c14_c15_c16_c17_c18, extending the naming convention by one segment per cycle.

3.2 Track 2 — Anchor preservation

The list continues to grow by exactly ten anchors per cycle as the previous cycle's own artifacts join the read-only anchor set.

Cycle	Anchor count	all_match	n_diff	Target
17	196	true	0	≥195
18	206	true	0	≥205
19	216	true	0	≥215

Cycle	Anchor count	all_match	n_diff	Target
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Every locked script remained byte-identical across all three cycles, including the Cycle 7 probe module (now a thirteen-cycle chain), the Cycle 12 probe module, and every earlier delivered verdict from Cycles 4 through 18.

3.3 Track 3 — Verdict emission

Each cycle emits its verdict at `data/v3/deliveries/31a164f845f8e27e/cycle<N>/verdict.json`:

Cycle	Verdict	On-disk SHA	cycles_since_last_operator_input
17	V3_SPINE_C17_ HEARTBEAT_pending_ operator	pinned by c18 backref	13
18	V3_SPINE_C18_ HEARTBEAT_pending_ operator	95a96f9561ef9de4e27fc314e798a7a7786ea13973049111a1f02e80f15715d5	14
19	V3_SPINE_C19_ HEARTBEAT_pending_ operator	1485f281acb42e3f13d50ee1001b8f1b0be14e733f1b122da566e2390ada6bfd	15

Each verdict carries `blocked_on_operator=true`, `cadence_mode=heartbeat`, a `prior_cycles` list of the fifteen prior cycles (Cycle 19's ends ...c16, c17, c18), the previous cycle's `c<N-1>_backref_sha` resolved from disk at emit time, and the two operator-facing WAV pins with `operator_ear_pending_fd6` status (Method A `cc919559b4508b6b...`; Method B `f40796be982998b0...`). The three-way rubric-v2 chain is byte-equal at each emission.

3.4 Track 4 — Housekeeping

Each cycle emits eight ledger events in strict order under a single cycle-scoped `run_id` (`run-2026-09-02T<HH>0000Z`, `HH` = 21 for Cycle 19; the earlier two cycles use `HH` consistent with their emission times):

1. `M-V3-SPINE-1/torch213-reproduce-probe-c<N>-completed`
2. `M-V3-SPINE-1/anchor-preservation-pre-c<N>-verified`
3. `M-V3-SPINE-1/anchor-preservation-post-c<N>-verified`
4. `M-V3-SPINE-1/verdict-c<N>-emitted (status=action_required)`
5. `M-INGEST-1/egress-probe-cycle<N>`
6. `_plan/register-c<N>-v3-spine-sub-leaves`
7. `_infra/adopt-cycle<N>-tests`
8. `_archive/cycle-<N>-scratch` (timestamped `ts+1s` after the physical move)

The egress probe row continues to record HTTP 429 + `tv_embedded` on every attempt. `promise_check` returned zero errors at every cycle.

3.5 Test discipline

Each cycle adds `tests/test_v3_spine_c<N>.py` with the same twelve-case shape and re-runs prior heartbeat suites plus the generic verdict-walker invariant. Cumulative sanity-floor tallies:

Cycle	New cases	Regression suites re-run	Total green
17	12	c16 through c9 + generic	104/104
18	12	c17 through c9 + generic	116/116
19	12	c18 through c9 + generic	128/128

The auditor independently re-ran the newest cycle’s suite and the generic invariant at each cycle under the mandated environment; all runs passed live.

4. Cycle-specific notes

Cycle 17 — ninth consecutive heartbeat. Anchor list grows to 196 (+10). No new observations.

Cycle 18 — tenth consecutive heartbeat. Anchor list grows to 206 (+10). No new observations; audit was 0 CRITICAL / 0 MODERATE / 1 MINOR (log-only orphan-report artifact, matching the pre-Cycle 4 pattern that had persisted since the pivot).

Cycle 19 — eleventh consecutive heartbeat. Anchor list grows to 216 (+10). The Cycle 19 audit was 0 CRITICAL / 0 MODERATE / 1 MINOR (elevated WARN growth trajectory) plus a repeat of the log-only orphan-report observation. The elevated-WARN observation is worth naming in detail because it is the first cycle in this arc to depart from the previously-converged pattern:

- Prior three cycles had shown `promise_check` warning-count deltas of +2, +2, +3 versus the previous cycle.
- Cycle 19’s on-disk count was 2784, versus Cycle 18’s baseline of 2779 — a delta of +5.
- Worker’s self-reported count was 2783; live re-run returned 2784, a small bookkeeping variance of +1 versus the worker claim.
- The auditor characterized the composition as routine orphan-report artifact accretion (per the Cycle 14 auditor’s earlier categorization of this WARN class) plus one row from a longstanding pre-Cycle 4 policy artifact.
- Since Cycle 19 is a heartbeat-only cycle (no substantive artifact changed the WARN class), and `promise_check` still returned 0 ERROR, the observation is treated as trend-worthy but not blocking.

The Cycle 19 auditor’s forward guidance is that a Cycle 20 researcher should explicitly note the on-disk WARN count in its `work_output` and cite the +1..+2 envelope with the actual delta so the auditor does not need to reconcile. If Cycles 20 and 21 continue to run at +4/+5 rather than +1/+2, the observation may be promoted to MODERATE and the drift class characterized. The auditor emphasized that at the moment the trajectory is worth watching, not fixing.

5. Cumulative state at end of arc

M-V3-SPINE-1 has now sat in `blocked_on_operator` state for fifteen cycles since Cycle 5 without regression on any load-bearing invariant. Fifteen `LANDS_pending_operator` verdicts sit on disk (Cycles 4 and 5 as substantive-cycle landings, Cycle 6 as the two-track close-out, Cycle 7 as the last three-track cycle, Cycle 8 as the moderate-fix cycle, and Cycles 9 through 19 as the eleven heartbeats), all byte-identical to their emission-time hashes.

The two operator-facing A/B candidates remain unchanged: Method A at `data/v3/deliveries/31a164f845f8e27e/section/full_reconstruction_operator_section.wav` (SHA `cc919559b4508b6b...`) and Method B at `data/v3_spine/rc7_v2_v3_paths/rc7_v2_v3_paths_full_reconstruction.wav` (SHA `f40796be982998b0...`).

The three operator-facing decisions queued at the end of Cycles 4–6 remain queued: ear verdict on either A/B pair, approval to execute torch-213 Mode 2, and the canonicity choice between the two mix chains. Downstream milestones M-V3-FOCUS-1, M-V3-CORPUS-1, M-V3-RULES-1, M-V3-EAR-1, and M-V3-GEN-1 remain frozen.

Three quantitative observations describe the heartbeat regime after this arc:

- **Fifteen consecutive clean audit cycles (Cycles 5-19)** with roughly a hundred and twenty-five cumulative live SHA spot-checks resolved on disk and zero fabrications detected.
- **Thirteen-cycle byte-identical venv manifest** (Cycles 7-19; roughly 34,697 files hashed identically at each pass), confirming that the c3-era torch 2.13.0+cpu attribution candidate remains reproducible in Mode 1 without environment drift accumulating during the wait.
- **Steady-state +10 anchors per cycle** confirmed across eight consecutive heartbeat cycles at this cadence, projecting to about 226 anchors at a hypothetical Cycle 20 heartbeat.

6. Conclusions

The Cycles 17–19 arc continues the campaign’s steady-state mode without incident. Eleven consecutive heartbeats have now landed under the Cycle 8 policy, three of them in this arc, and the fifteen-cycle clean-audit streak proves that the discipline invariants encoded in the policy remain automatically enforced. The single novel observation of the arc — a Cycle 19 warning-count growth delta slightly above the previously-converged envelope — is a routine bookkeeping trend flagged for watching, not a discipline defect. Every load-bearing chain (rubric hash, cadence policy hash, venv byte-identity, anchor preservation, append-only verdicts, Mode-2 lock) remains byte-equal cycle over cycle.

The unresolved work continues to be operator judgment, not code. Any single one of the three queued operator decisions arriving in a subsequent cycle’s `live_guidance` immediately breaks the glass and reopens substantive work; until then, the heartbeat cadence continues linearly and the milestone remains available for immediate positive resolution as soon as an ear verdict lands.

Appendix: Implementation Details

A.1 Delivered artifacts by cycle

Cycle 17, verdict at `data/v3/deliveries/31a164f845f8e27e/cycle17/verdict.json`; working artifacts under `data/v3_spine/cycle17/`.

Cycle 18, verdict at `data/v3/deliveries/31a164f845f8e27e/cycle18/verdict.json` (SHA 95a96f9561ef9de4e27fc314e798); working artifacts under `data/v3_spine/cycle18/`.

Cycle 19, verdict at `data/v3/deliveries/31a164f845f8e27e/cycle19/verdict.json` (SHA 1485f281acb42e3f13d50ee1001b); working artifacts under `data/v3_spine/cycle19/` including torch probe JSON, anchor pre/post/diff, and driver-script outputs.

Cycle-scoped reports at `docs/v3_spine_report_cycle17.md`, `docs/v3_spine_report_cycle18.md`, `docs/v3_spine_report_cycle19.md`.

A.2 Scripts

Per cycle: `scripts/v3_spine/torch213_reproduce_probe_c<N>.py`, `scripts/v3_spine/anchor_preservation_c<N>.py`, `scripts/v3_spine/verdict_c<N>.py`. One-shot ledger emitters born under `tools/stale/cycle<N>_v3_spine_scratch/` per cycle.

Cycle 7 probe module locked at `b54adadd1f4f04028d4d931710cc02b722c82ebee027728a97adfa8b20c9af3d` (thirteen-cycle byte-identical chain through Cycle 19). Continuing locked: `scripts/palette_render/render_stem.py`, `scripts/v3_spine/mix_match_operator_section.py`, `scripts/v3_spine/rc7_v2_rerun_v3_paths.py`, `scripts/recreate_v2/rc7_v2_rerun.py`, `scripts/v3_spine/rc7_mix_balance.py`, plus every earlier per-cycle probe and verdict driver.

A.3 Tests

`tests/test_v3_spine_c17.py`, `tests/test_v3_spine_c18.py`, `tests/test_v3_spine_c19.py` — twelve cases each; `tests/test_verdict_sha_fields_resolve_on_disk.py` — eight generic-invariant cases. Full sanity-floor regression re-run at every cycle. Cycles 17/18/19 tallies: 104/104, 116/116, 128/128 green.

A.4 Integrity chains

Three-way rubric-v2 chain: `docs/v3_spine_rubric_v2.md` SHA `c49db5a12e955f26c001165ad6e8f9d191bc26bfd96e24c1b1` == `data/v3_spine/rubric_hash_v2.txt` content == each cycle's verdict.`rubric_hash_v2`. Cadence policy chain: `docs/wait_on_operator_cadence_policy.md` SHA `0be540365c8c03ad38a15478fbad0fe32bf5ea4118e33ef3eed` == its pinned hash file.

Venv directory-manifest byte-identity across thirteen cycles (Cycles 7 through 19): `a86205175728d58f0a96ad02fc1ab1ac9e35f06c5ed568a960ed1ff261f83a74`.

Verdict backref chain: each cycle pins the previous verdict SHA; Cycle 19 pins Cycle 18's `95a96f9561ef9de4...15715d5`; auditor re-verifies on-disk at each pass.

A.5 Ledger schedule

Each cycle emits eight events under `run_id = run-2026-09-02T<HH>0000Z`, timestamps `T<HH>:00:00Z` for the first seven events and `T<HH>:00:01Z` for the post-move archive row (Cycle 19: `21:00:00Z` × 7 + `21:00:01Z`).

A.6 Environment pins

`PYTHONHASHSEED=0`; `SOURCE_DATE_EPOCH=1756463424`; `TZ=UTC`; `LC_ALL=C.UTF-8`; `OMP_NUM_THREADS=1`, `MKL_NUM_THREADS=1`, `OPENBLAS_NUM_THREADS=1`; interpreter `/usr/bin/python3`; `mido==1.3.3`; SoundFont SHA `74594e8f...1cb0`; MuScriptor model SHA `ac80adbd...7fb97ec`.

A.7 Anchor growth profile

Cycle 16 → Cycle 17: 186 → 196 (+10); Cycle 17 → Cycle 18: 196 → 206 (+10); Cycle 18 → Cycle 19: 206 → 216 (+10). Growth per cycle corresponds to the ten heartbeat-cycle-specific artifacts that lock at cycle close (verdict, torch-probe JSON, three anchor-preservation JSONs, three driver scripts, twelve-case test file, cycle-scoped report document).

A.8 promise_check WARN trajectory

Recent per-cycle WARN counts on disk: Cycle 15 = 2774; Cycle 16 = 2774; Cycle 17 = 2777; Cycle 18 = 2779; Cycle 19 = 2784 (+5 versus Cycle 18). ERROR count stayed at zero throughout. Cycle 19 audit flagged the trajectory as MINOR log-only, noting a small bookkeeping variance (+1) between the worker's self-reported count (2783) and the auditor's live re-run (2784), and recommended that the Cycle 20 researcher cite the actual on-disk count with the actual delta so no reconciliation is required.

A.9 Cumulative audit statistics

Fifteen consecutive clean audit cycles (Cycles 5–19); roughly one hundred and twenty-five cumulative live SHA spot-checks across the arc with zero fabrications detected; 0 CRITICAL and 0 MODERATE findings surfaced anywhere in the eleven-cycle heartbeat era. The two MINOR observations from Cycle 19 (elevated WARN growth trajectory; longstanding orphan-report artifact) are logged only and do not block.

A.10 Source sessions

Cycle	Researcher	Worker	Auditor
17	f2a67751-49f1-4ef9-b519-fedbf358f2f7	1fa7d650-cd13-4bc3-985e-1161f73c6c42	96a71a2e-e281-453e-a05e-89c2b2bf39bb
18	74ba7f27-bdd7-4903-a0ea-c34bd43b09e3	9dedf895-d9e5-4402-8fd0-a215aa02b9d7	43520aab-cb2b-4fea-9785-a33d798b15a5
19	b83da495-bedc-4633-8dbd-0f91cba410f7	285b5cad-8c18-4f34-af30-ce3e6b156520	a00f1e19-96f1-46a0-8554-d04ca26198c7